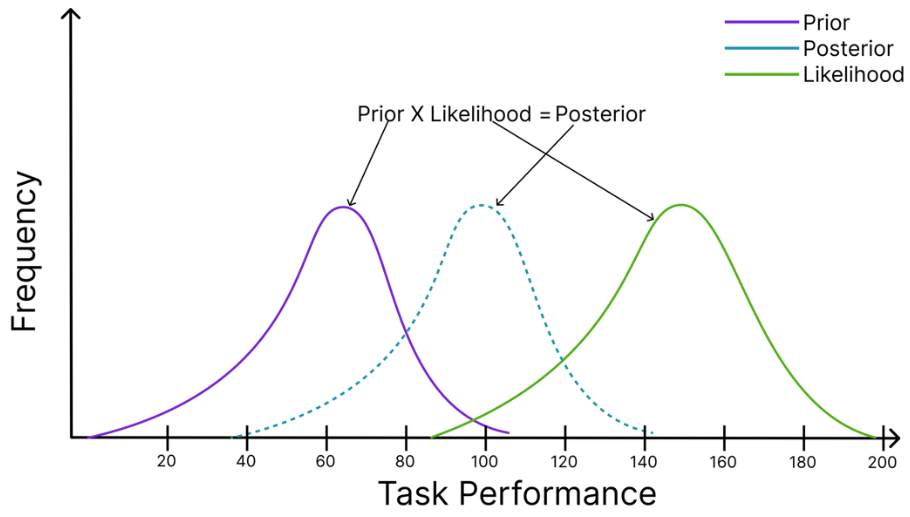
Modelling Approach

Bayesian Modelling Approach

Adopting an approach from **Buchel et al.(2014)** and **Strube et al. (2023)**,we Modeled objective performance as Bayesian inference:

Prior Expectation + Task Evidence → Predicted Performance

Goal: Investigated how AI placebo can be explained in terms of cognitive mechanisms underlying prior performance



Model Parameterisation:

Key latent parameters estimated were:

- p_shift: Represents a *shift in the mean* of prior expectation (added to stated expectancy).

- log_var_scale: Represents a *change in the log variance (inverse of precision)* of prior expectation.

Three models with different free parameters were created:

- **Shift Model:** Estimates by shift prior mean
- **Precision Model:** Changing precision of prior expectations
- **Combined Model:** Has both the parameters free

Model Testing

Parameter Estimation: Markov Chain Monte Carlo (MCMC) sampling was used to approximate the joint posterior distribution of all model parameters for each of the three specified models.

Implementation: The analysis utilized the PyMC library (pm.sample) and its default18 high-efficiency sampler, the No-U-Turn Sampler (NUTS).

Sampling Settings:

- 4 independent chains, each generating
- 2500 posterior samples following an
- initial 2500 tuning steps.

The target_accept parameter= 0.98.
A fixed random seed was used for reproducibility.

Log-likelihood values for observed variables were computed and stored.

Model Comparison:

Calculated:

- Leave-One-Out cross-validation (LOO)
- Widely Applicable Information Criterion (WAIC)

The models are ranked based on their **LOO Scores.**

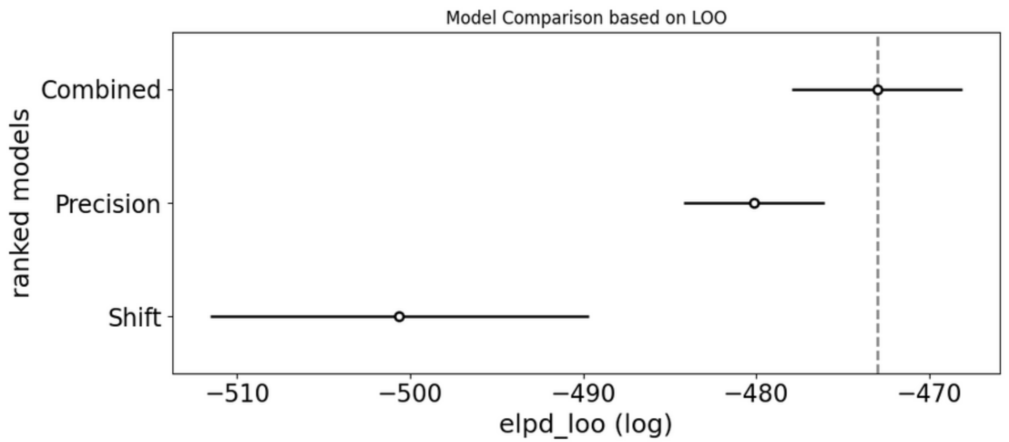
Lower scores indicate better out-of-sample predictive accuracy.

Correlation:

Pearson Correlation Coefficient was used to asses the relationship between individual differences in the estimated changes of the model parameters and the individual differences in observed objective task performance

Results

Hypotheses 1:



Best Fitting Model (H1 Supported):

- The Combined Model provided the best fit.
- Suggests placebo involves modulating both expected performance level and certainty.

Group-Level Cognitive Parameters (H2 Not Supported):

- No statistically significant *average change* in estimated prior mean shift or log prior variance at the group level.
- Mirrors original finding of no group effect on objective performance.

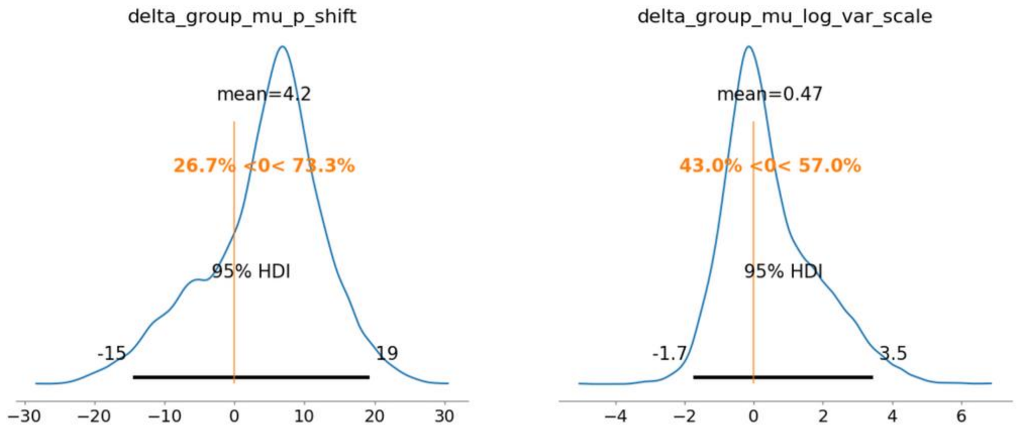
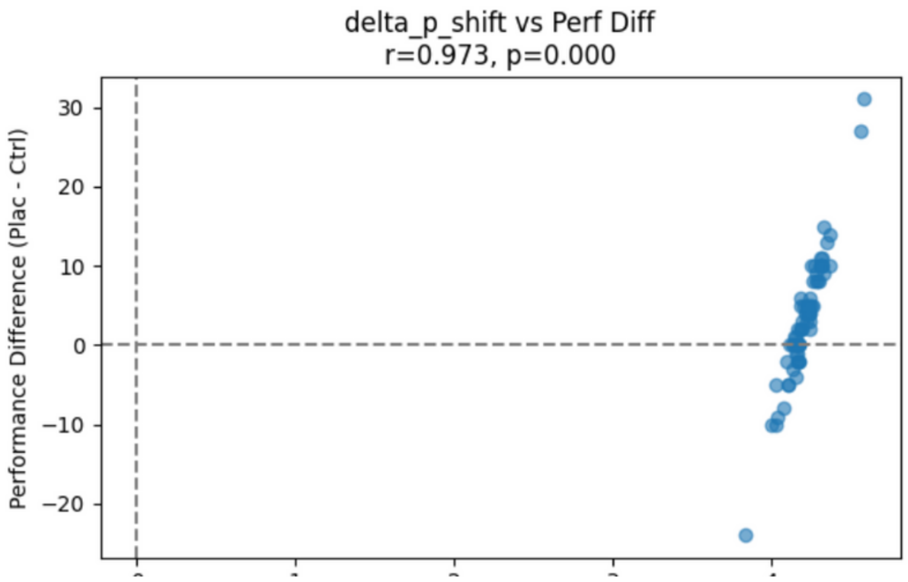


Figure 2.1: Posterior Distribution of group Mean Differences

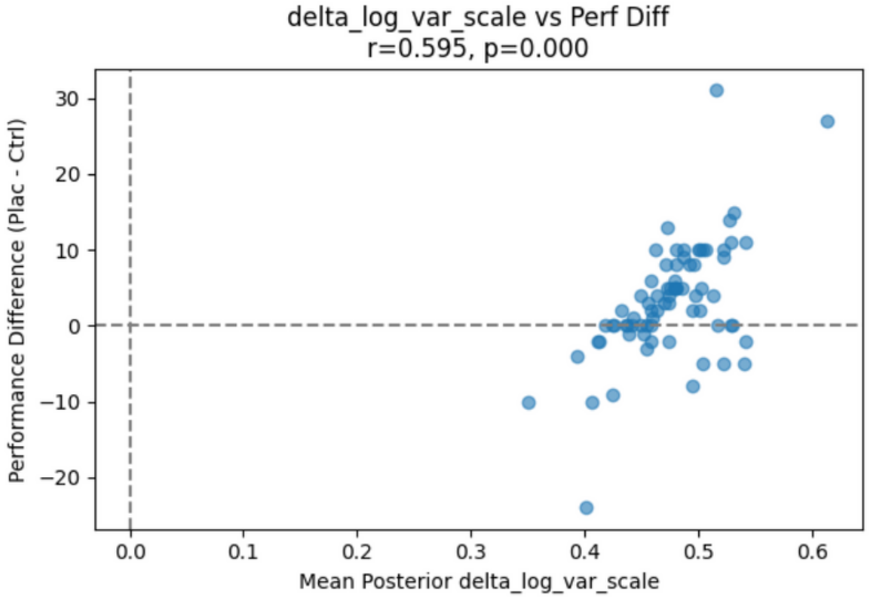
Results

Individual-Level Correlations (H3 - Strong Findings):

Prior Mean Shift (H3a Supported):
Strong positive correlation ($r=0.973$, $p < 0.0001$) between individual delta_p_shift and objective performance difference. (More shift = more gain).



Prior Precision (H3b Opposite Finding): Moderate positive correlation ($r=0.595$, $p < 0.0001$) between individual delta_log_var_scale and performance difference. (More variance/less precision = more gain - an unexpected result).



Discussion

- AI placebo effect likely involves changes in *mean* (level) and *precision* (certainty) of prior performance beliefs.
- No average group change in cognitive parameters; however, individual *increases in prior mean expectation* strongly predicted individual *objective performance gains*.
- Placebo's impact is individual-specific, tied to prior mean shifts. Unexpectedly, *decreased prior precision* (more variance) also correlated with performance gains.

Contributions:

- Quantified individual-specific latent cognitive parameter changes.
- Explained *why* individual expectations link to performance (via prior mean shifts).
- Highlighted importance of studying individual differences in HCI placebo effects.

Future Directions

- Investigate the counterintuitive precision effect.
- Replicate & generalize findings across diverse tasks, AI manipulations, and populations.
- Extend models: incorporate moderators (e.g., personality), subjective ratings.
- Design studies with direct measures of confidence/certainty